

GODWIN EDGAR OPUKA

Full-Stack Solutions Architect | AI/ML Engineer | Financial Engineering | MLOps | Graph Data Science

Location: Nairobi, Kenya	Phone: +254 715 849 117
Availability: Open to Full-Time, Remote & Hybrid Roles	LinkedIn: linkedin.com/in/godwin-edgar-opuka

Professional Summary

Full-stack Solutions Architect, AI/ML Engineer, and Financial Engineer with strong expertise in designing, building, and delivering end-to-end intelligent systems — from backend data pipelines and API layers through to frontend interfaces and graph-powered analytics. Proven experience owning the full development lifecycle of enterprise-scale digital platforms, architecting microservices, GraphQL/REST APIs, React-based frontends, and AI-driven analytics solutions aligned to MACH architecture principles. Currently serving as Solutions Architect for a Digital Business Platform (Twin Graph Manager) and a Digital Intelligence & Analytics solution. Adept at bridging architecture, engineering, and data science to deliver production-grade platforms.

Core Competencies

Full-Stack Development	Solutions Architecture	Machine Learning & MLOps
Graph Data Science	API & Microservices Design	Data Engineering
AI Agent Development	Business Intelligence	Financial Engineering

Technical Stack

Domain	Technologies
Languages & Analytics AI / ML	Python, R, Spark, SQL Machine Learning Pipelines, Feature Engineering, Model Deployment, AI Agents, CI/CD for ML

Domain	Technologies
Graph Data Science	Neo4j, Graph Modeling, Graph Algorithms, Network Analysis
Data & BI	Microsoft Fabric, Power BI, Azure Data Factory
API Development	REST APIs, GraphQL APIs, Microservices Architecture
Web Frameworks	React.js, Next.js, Node.js
Enterprise Architecture	MACH Architecture, Ardoq, Abacus
DevOps & Delivery	Azure DevOps, Agile/Scrum, CI/CD Pipelines

Professional Experience

DigitalQatalyst (Contract)

Jun 2025 – Present

Solutions Architect & BI Lead *Promoted Sep 2025*

Owning end-to-end solution architecture and full-stack development across two flagship digital business platforms — from system design and backend services through to frontend delivery and analytics integration.

Solutions Architecture — Digital Business Platform (Twin Graph Manager):

- Architecting the Twin Graph Manager, a graph-powered digital business platform that models and manages enterprise digital twins using Neo4j and graph data science.
- Designing and building the full-stack platform architecture: Neo4j graph backend, GraphQL/REST API layer, Redux-based state management, and React frontend with component-driven UI.
- Implementing microservices patterns and MACH-aligned architecture to ensure platform scalability, composability, and extensibility.
- Hands-on development across the stack — writing backend resolvers, designing data models, building reusable frontend components, and configuring CI/CD pipelines.

Solutions Architecture — Digital Intelligence & Analytics:

- Architecting a full-stack Digital Intelligence and Analytics solution delivering descriptive, diagnostic, predictive, prescriptive, and deep intelligence capabilities.

- Designing and implementing analytics data pipelines, ML model serving endpoints, and interactive React-based dashboards powered by graph-based relationship analytics.
- Building API-first data services (REST & GraphQL) with governance standards for enterprise-grade consumption.

Business Intelligence & Enterprise Architecture:

- Built graph-based intelligence solutions using Neo4j to support relationship-centric analytics and decision intelligence.
- Oversaw enterprise architecture modeling using Ardoq and Abacus to align business capabilities with digital platforms.
- Directed deployment of AI agents for business process automation across multiple domains.

Key Achievements:

- **Enterprise Architecture IP Development** — Architected the development of company IP EA solution to automate digital transformation through facilitation of reliable specifications via graph-powered analysis of digital capabilities and digital initiatives.
- **Dashboard Modernization & Migration** — Orchestrated the successful transition, design, and deployment of dashboards from Power BI to React-based solutions for seamless integration into digital platforms.

Full-Stack Data Developer | AI/ML Engineer *Jun 2025 – Sep 2025*

- Engineered end-to-end data pipelines using Python, R, Spark, and Microsoft Fabric.
- Developed and deployed ML-powered services with REST and GraphQL API endpoints, including frontend integration layers in React.
- Designed feature engineering pipelines for predictive analytics and AI-driven products.
- Built and maintained CI/CD pipelines for data, ML, and application workloads using Azure DevOps.

Data Analyst

Freelance | Apr 2021 – May 2025

- Delivered quantitative and qualitative analytical solutions across finance and business domains.
- Conducted financial statement analysis and hypothesis testing using robust statistical methods.
- Generated actionable insights through advanced data modeling and research methodologies.

Personal Projects

Market Signal Graph

Jan 2025 – Present

A next-generation financial research tool that combines vector search (pgvector on PostgreSQL) with graph data science (Neo4j) to uncover hidden market signals and surface non-obvious relationships in financial data.

High-Performance Derivatives Pricing Engine

Jan 2025 – Present

Implemented mathematical models in Rust with a FastAPI backend to compute derivative prices at high performance, bridging quantitative finance theory with production-grade engineering.

Internships

Role	Organisation	Period
Claims Analyst	Madison Group Kenya	Jul 2022 – Nov 2022
Asset Management Analyst	Nairobi County Government	Jan 2022 – Mar 2022
Business Development Associate	Favitech Solutions Ltd	Oct 2020 – Mar 2021

Education

MSc Financial Engineering

WorldQuant University | Jul 2023 – Apr 2026 (*Expected*)

A rigorous, fully online quantitative finance program with a curriculum spanning mathematical modeling, computational finance, and applied machine learning — built for practitioners bridging data science and financial markets.

Coursework Completed:

Course	Key Skills & Concepts
Financial Data	Financial data sourcing, cleaning, time-series analysis, and signal extraction

Course	Key Skills & Concepts
Financial Markets	Market microstructure, asset classes, pricing dynamics, and market behavior
Derivative Pricing	Options theory, Black-Scholes, Monte Carlo simulation, and Greeks
Financial Econometrics	Regression analysis, volatility modeling (GARCH), and statistical inference
Stochastic Modeling	Brownian motion, Itô calculus, SDEs, and stochastic processes in finance
Machine Learning in Finance	Supervised/unsupervised ML, feature engineering, and alpha signal generation
Deep Learning for Finance	Neural networks, LSTM models, and deep learning for financial prediction
Delivering Impactful Presentations	Quantitative research communication and stakeholder-facing financial reporting

Currently Enrolled: - Portfolio Management — Modern Portfolio Theory, optimization, factor models, and risk-adjusted performance

Remaining: - Risk Management | Capstone Project

BSc Financial Engineering

Jomo Kenyatta University of Agriculture and Technology (JKUAT)
| Sep 2017 – Dec 2021

A mathematically intensive undergraduate program grounding financial engineering in rigorous quantitative theory — from pure mathematics through to advanced stochastic and financial modeling.

Coursework Covered:

Domain	Subjects
Mathematical Foundations	Differential Calculus, Integral Calculus, Discrete Mathematics, Linear Algebra, Complex Analysis, Real Analysis
Probability & Statistics	Probability Theory (Introductory to Advanced), Statistical Inference, Numerical Analysis

Domain	Subjects
Advanced Mathematics	Stochastic Calculus, Extreme Value Theory, Numerical Methods
Financial Theory & Modeling	Financial Theory, Derivative Pricing, Game Theory

References

Name	Title & Organisation	Contact
Ibrahim Abdi	Systems Engineer – FS Planning, Safaricom PLC	+254 719 310 268
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